

Review-2

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Divisibility

Shapes, Counting, Pattern

Exponents

Logic, Chicken Rabbit Problem

Multiple and Factors

Fractions

Decimals

Divisibilit y

Ways to recognize divisibility by 2, 5

A number is divisible by 2 if and only if its unit digit is even

408, 602, 704 Yes

985, 513, 437 No

A number is divisible by 5 if and only if its unit digit is 0 or 5

405, 630, 575 Yes 231,378, 5936 No

Ways to recognize divisibility by 3

If the sum of digits of the number is divisible by 3, then the number itself is divisible by 3.

510408, 6170, 54201, 4382, 2121, 25056

51408, 54201, 2121, 25056

Ways to recognize divisibility by 4

A number is divisible by 4 if and only if the number formed by its last two digits is divisible by 4.

A number that ends in 00,04,08,...,88,92,96 is divisible by 4

$$ABCDE \div 4 = (ABC \times 100 + DE) \div 4 = 25 \times ABC + DE \div 4$$

**376, 552, 991, 6036, 975, 805,
1416, 2584, 4072, 3700, 5556, 4137**

**376, 552, 6036, 1416,
2584, 4072, 3700, 5556**

Rule 1

Ways to recognize divisibility by 7

Starting from the unit digit, divide the number into segments of three digits each. If the difference between the sum of the odd-numbered segments and the sum of the even-numbered segments is divisible by 7, then the number is **divisible by 7**.

2093

$$93 - 2 = 91$$

$$91 \div 7 = 13$$

2093 is divisible by 7

8638

$$638 - 8 = 630$$

$$630 \div 7 = 90$$

8638 is divisible by 7

Rule 2

Ways to recognize divisibility by 7

Take the last digit of the number

Double it.

Subtract the result from the rest of the number .

Repeat the process if necessary.

For example, consider the number 588

- Last digit 8
 - Double it: $8 \times 2 = 16$
 - Subtract from the rest: $58 - 16 = 42$
 - Since 42 is divisible by 7, then 588 is divisible by 7
-

Find the number that is divisible by 7

9015, 3366, 6713, 2927, 317849, 246813

$$713 - 6 = 707$$

$$707 \div 7 = 101$$

Thus, 6713 is
divisible by 7

$$849 - 317 = 532$$

$$53 - 2 \times 2 = 49$$

$$49 \div 7 = 7$$

Thus, 317849 is
divisible by 7

$$813 - 246 = 567$$

$$56 - 2 \times 7 = 42$$

$$42 \div 7 = 6$$

Thus, 246813 is
divisible by 7

Ways to recognize divisibility by 8

If the last three digits of a number is divisible by 8, then the number is **divisible by 8**.

$$ABCDEF \div 8$$

$$= (ABC000 + DEF) \div 8$$

$$= ABC000 \div 8 + DEF \div 8$$

$$= ABC \times 1000 \div 8 + DEF \div 8$$

$$= ABC \times 125 + DEF \div 8$$

Find the number that is divisible by 8

132000, 784165, 52458, 23341, 12432

132000, 12432

Ways to recognize divisibility by 9

If the sum of digits of the number is divisible by 9, then the number itself is divisible by 9.

5904, 29960, 300627, 75753, 26432, 38247

5904, 300627, 75753

Rule 1**Ways to recognize divisibility by 11**

If the difference of the sum of digits in the even place values and the sum of the digits in the odd place values is 0 or a multiple of 11, the number is **divisible by 11**

135795, 123456, 258016, 234768

135795, $3+7+5-(1+5+9)=0$, Yes

123456, $2+4+6-(1+3+5)=3$, No

258016, $5+0+6-(2+8+1)=0$, Yes

234765, $3+7+8-(2+4+6)=6$, No

Rule 2**Ways to recognize divisibility by 11**

Starting from the unit digit, divide the number into segments of three digits each. If the difference between the sum of the odd-numbered segments and the sum of the even-numbered segments is a multiple of 11, then the number is **divisible by 11**.

246840, 654321, 987646, 314159

$$840 - 246 = 594$$

$$5 + 4 - 9 = 0$$

$$0 \div 11 = 0$$

Thus, 246840 is
divisible by 11.

$$987 - 646 = 341$$

$$3 + 1 - 4 = 0$$

$$0 \div 11 = 0$$

Thus, 987646
divisible by 11.

Rule 1**Ways to recognize divisibility by 13**

Starting from the unit digit, divide the number into segments of three digits each. If the difference between the sum of the odd-numbered segments and the sum of the even-numbered segments is a multiple of 13, then the number is **divisible by 13**.

4420

$$420 - 4 = 416$$

$$416 \div 13 = 32$$

4420 is divisible by 13

21710

$$710 - 21 = 689$$

$$689 \div 13 = 53$$

21710 is divisible by 13

Rule 2**Ways to recognize divisibility by 13**

Take the last digit of the number.

Multiply it with 4.

Add the result to the rest of the number .

Repeat the process if necessary.

For example, consider the number 741

- Last digit 1
 - Multiple it with 4: $1 \times 4 = 4$
 - Add to from the rest: $74 + 4 = 78$
 - Since 78 is divisible by 13, then 741 is divisible by 13
-

Find the numbers that are divisible by 13

8015, 5138, 17810, 15234, 246805

$$810-17=793$$

$$79+3\times 4=91$$

$$91\div 13=7$$

Thus, 17810 is
divisible by 13.

$$805-246=559$$

$$55+9\times 4=91$$

$$91\div 13=7$$

Thus, 246805 is
divisible by 13.

Ways to recognize divisibility by 16

If the last four digits of a number is a multiple of 16, then the number is a divisible by 16.

24692, 65424, 543206, 89736, 32752

$$5424 \div 16 = 339$$

Thus, 65424 is
divisible by 16.

$$2752 \div 16 = 172$$

Thus, 32752 is
divisible by 16.

Ways to recognize divisibility by 17

Take the last digit of the number.

Multiply it with 5.

Subtract the result from the rest of the number .

Repeat the process if necessary .

For example, consider the number 476

- Last digit 6
 - Multiple it with 5: $6 \times 5 = 30$
 - Subtract it to from the rest: $47 - 30 = 17$
 - Since 17 is divisible by 17, then 476 is divisible by 17.
-

Find the numbers that are divisible by 17

4821, 2346, 87654, 3059, 45662

$$234 - 6 \times 5 = 204$$

$$204 \div 17 = 12$$

Thus, 2346 is
divisible by 17.

$$4566 - 2 \times 5 = 4556$$

$$455 - 6 \times 5 = 425$$

$$42 - 5 \times 5 = 17$$

$$17 \div 17 = 1$$

Thus, 45662 is
divisible by 17.

Ways to recognize divisibility by 19

Take the last digit of the number.

Multiply it with 2.

Add the result to rest of the number .

Repeat the process if necessary .

For example, consider the number 361

- Last digit 1
 - Multiple it with 2: $1 \times 2 = 2$
 - Subtract it to the rest: $36 + 2 = 38$
 - Since 38 is divisible by 19, then 361 is divisible by 19.
-

Find the numbers that are divisible by 19

2047, 3743, 48015, 654321, 45638

$$374 + 3 \times 2 = 380$$

$$380 \div 19 = 20$$

Thus, 3743 is
divisible by 19.

$$4563 + 8 \times 2 = 4579$$

$$457 + 9 \times 2 = 475$$

$$47 + 5 \times 2 = 57$$

$$57 \div 19 = 3$$

Thus, 45638 is
divisible by 19.

Ways to recognize divisibility by 25

A number is divisible by 25 if and only if the number formed by its last two digits is divisible by 25.

A number that ends in **00,25,50, 75** is divisible by 25

$$ABCDE \div 25 = (ABC \times 100 + DE) \div 25 = 4 \times ABC + DE \div 25$$

375, 517, 990, 6035, 625, 800,
1550, 2360, 4005, 3300, 5502, 3505

375, 625, 800, 1150, 3300

Ways to recognize divisibility by 99

"two-digit splitting and summing" method

For example, consider the number 10|09|80

- Split the number into two-digit groups from right to left:
 - Sum the two-digit groups:
 $80+9+10=99$
 - Check divisibility by 99
 $99\div 99=1$
-

Find the numbers that are divisible by 99

19503, 724162, 52152, 424611, 62418

19503, 424611

Ways to recognize divisibility by 999

“Three-digit splitting and summing” method

For example, consider the number 142|857

- Split the number into three-digit groups from right to left:
 - Sum the three-digit groups:
 $857 + 142 = 999$
 - Check divisibility by 999
 $999 \div 999 = 1$
-

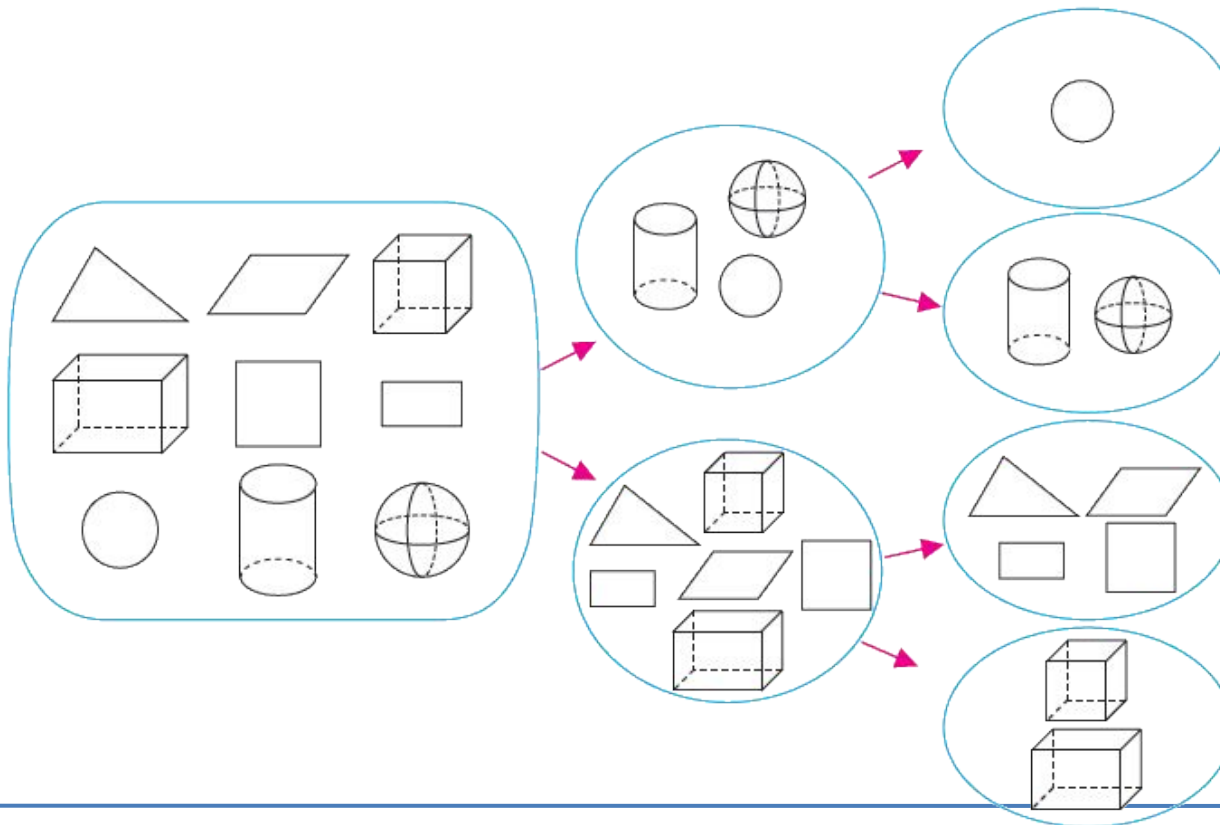
Find the numbers that are divisible by 999

37501, 45954, 522659, 62411, 307692

45954, 307692

Shapes, Counting, Pattern

Classify the graphics



Review

Classify triangles according to angles

Acute-angled triangle: each angle is less than 90 degree

Right-angled triangle: one angle is 90 degree

Obtuse-angled triangle: one angle is more than 90 degree

Review

Classify triangles according to sides

Scalene triangle: A triangle with no two sides of equal length

Isosceles triangle: A triangle with at least two equal sides

Equal lateral triangle is a special case of isosceles triangle

Polygon Name	Number of Interior Angles	Sum of Interior Angles = $(n-2) \times 180^\circ$
Triangle	3	180°
Quadrilateral	4	360°
Pentagon	5	540°
Hexagon	6	720°
Septagon	7	900°
Octagon	8	1080°
Nonagon	9	1260°
Decagon	10	1440°

Conclusion

The sum of any two sides of a triangle is greater than the third side.

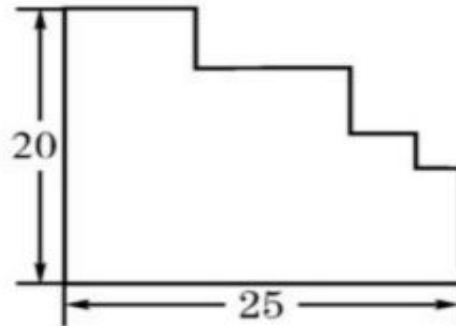
The difference of any two sides of a triangle is less than the third side.

Perimeter

1. The length around the edge of an object in one **complete** revolution is called the perimeter of that object.

2. The perimeter is the **sum** of all the line segments that form the shape

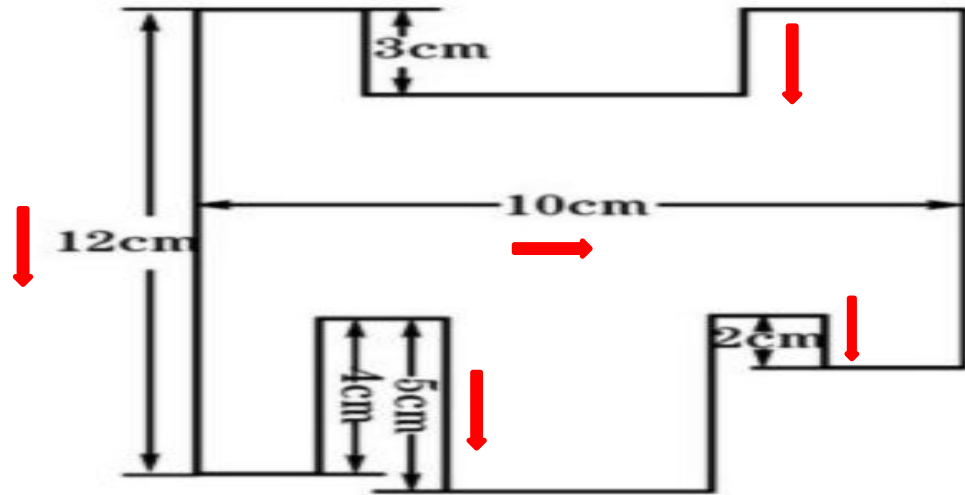
Find the perimeter



Solution

$$(20 + 25) \times 2 = 90.$$

Find the
perimeter



Solution

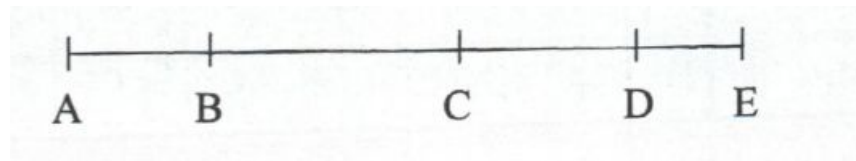
right: 10cm

down: $5+2+3+12=22(\text{cm})$

Total: $(10+22) \times 2 = 64(\text{cm})$

Example

How many segments of different length are there ?



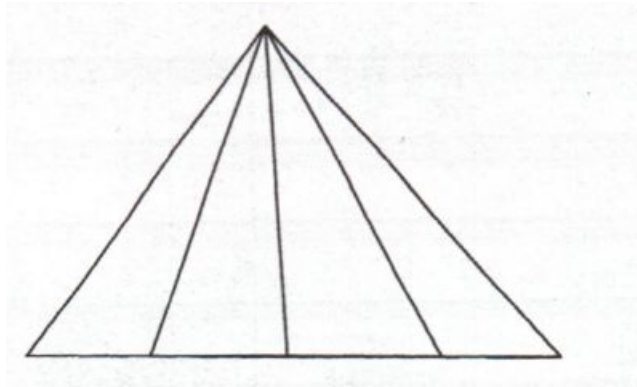
Solution:

AB	AC	AD	AE
BC	BD	BE	
CD	CE		
DE			

$$4 + 3 + 2 + 1 = 10$$

There are **10** lines of different lengths.

How many triangles are there altogether in the figure



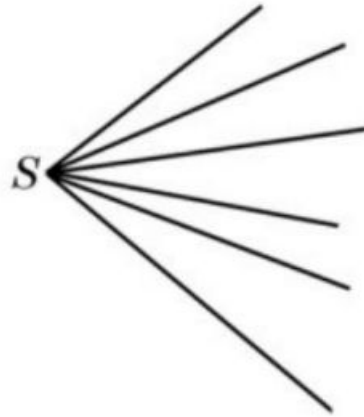
Same logic as counting segments

Solution

$$\begin{aligned} &1+2+3+4 \\ &=10 \end{aligned}$$

Example

The largest angle is less than 90 degrees. How many angles in the diagram are less than 90 degrees?



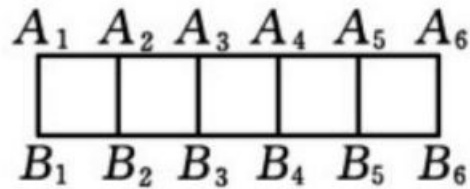
Same logic as counting segments

Solution

$$\begin{aligned} &1+2+3+4+5 \\ &=15 \end{aligned}$$

Example

How many rectangles are there altogether in the figure

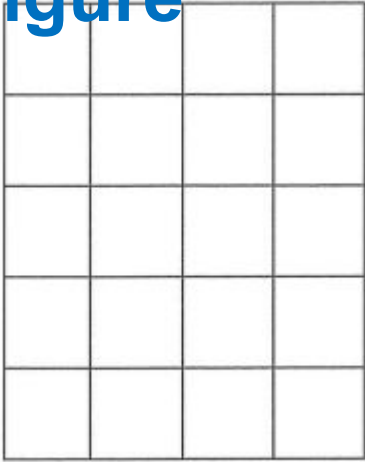


Same logic as counting segments

Solution

$$\begin{aligned} &1+2+3+4+5 \\ &=15 \end{aligned}$$

How many squares are there altogether in the figure



$$1 \text{ squares} = 5 \times 4 = 20$$

$$4 \text{ squares} = (5-1) \times (4-1) = 12$$

$$9 \text{ squares} = (5-2) \times (4-2) = 6$$

$$16 \text{ squares} = (5-3) \times (4-3) = 2$$

$$\text{total} = 20 + 12 + 6 + 2 = 40$$

Complete the number patterns

(a) 1, 2, 4, 8, 16, (32), (64), ...

(b) 3, 4, 5, 8, 7, 16, 9, 32, (11), (64), ...

(c) 0, 3, 8, 15, 24, (35), (48), 63, ...

(a) 1, 2, 4, 8, 16, 32, 64, ...

$\times 2$ $\times 2$ $\times 2$ $\times 2$ $\times 2$ $\times 2$

(b) 3, 4, 5, 8, 7, 16, 9, 32, 11, 64, ...

$\times 2$ $\times 2$ $\times 2$ $\times 2$

$+2$ $+2$ $+2$ $+2$

(c) 0, 3, 8, 15, 24, 35, 48, 63, ...

$+3$ $+5$ $+7$ $+9$ $+11$ $+13$ $+15$

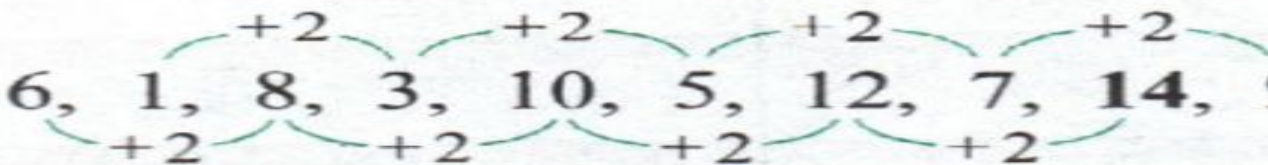
Complete the number patterns

(d) 6, 1, 8, 3, 10, 5, 12, 7, (14), (9), ...

(e) 2, 3, 5, (8), (12), 17, (23), ...

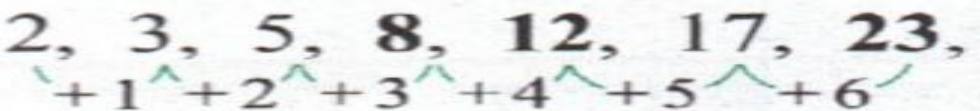
(f) 3, 5, 6, 10, (9), 15, (12), (20), ...

(d) 6, 1, 8, 3, 10, 5, 12, 7, 14, 9, ...



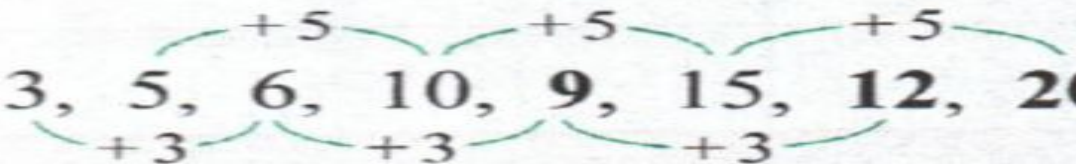
The diagram shows the sequence 6, 1, 8, 3, 10, 5, 12, 7, 14, 9, ... with green curved arrows indicating the steps: +2 (6 to 1), -2 (1 to 8), +2 (8 to 3), -2 (3 to 10), +2 (10 to 5), -2 (5 to 12), +2 (12 to 7), -2 (7 to 14), and +2 (14 to 9).

(e) 2, 3, 5, 8, 12, 17, 23, ...



The diagram shows the sequence 2, 3, 5, 8, 12, 17, 23, ... with green curved arrows indicating the steps: +1 (2 to 3), +2 (3 to 5), +3 (5 to 8), +4 (8 to 12), +5 (12 to 17), and +6 (17 to 23).

(f) 3, 5, 6, 10, 9, 15, 12, 20, ...



The diagram shows the sequence 3, 5, 6, 10, 9, 15, 12, 20, ... with green curved arrows indicating the steps: +2 (3 to 5), +3 (5 to 6), -3 (6 to 10), +3 (10 to 9), -3 (9 to 15), +3 (15 to 12), and -3 (12 to 20).

Exponent s

Basic Exponent Rules

- Multiplying with the same base

- $a^m \times a^n = a^{m+n}$

- Dividing with the same base

- $a^m \div a^n = a^{m-n}$

- Raising a power to a power

- $(a^m)^n = a^{m \cdot n}$

Practice

$$5^4 = x^2$$

$$x = 5^2 = 5 \times 5 = 25$$

$$6 \times 8 \times 6 \times 8 = x^2$$

$$x = 6 \times 8 = 48$$

$$3^2 \times 8^4 = x^2$$

$$x = 3 \times 64 = 192$$

Practice

Which is greater 5^{22} or 3^{33}

$$5^{22} = (5^2)^{11} = 25^{11}$$

$$3^{33} = (3^3)^{11} = 27^{11}$$

$$25^{11} < 27^{11} \quad \longrightarrow \quad 5^{22} < 3^{33}$$

Logic, Chicken Rabbit Problem

What's Logic ?

Logic is a branch of mathematics that relies more on strong reasoning skills rather than on computational capability

Techniques

Assume something to be true until it is proven otherwise

Make a systematic list to scrutinize the given information

Make a table or a few tables

Example

A football flew right onto Miss Cookson's back. 3 suspected culprits were called into the Discipline Master's room for investigation.

Shawn: I did not do it!

Andy: Not me!

James: Andy lied.

One of them told the truth. Who kicked the football?

If Shawn did it

	Lie	Truth
Shawn	Yes	
Andy		Yes
James	Yes	

If Andy did it

	Lie	Truth
Shawn		Yes
Andy	Yes	
James		Yes

If James did it

	Lie	Truth
Shawn		Yes
Andy		Yes
James	Yes	

Shawn kicked the football

$$\blacklozenge + \bigcirc = 54$$

$$\blacklozenge = \bigcirc + \bigcirc + \bigcirc + \bigcirc + \bigcirc$$

Find the values of each $\blacklozenge$ and each $\bigcirc$.

Solution: $6 \bigcirc = 54$

$$\bigcirc = 54 \div 6 = 9$$

$$\blacklozenge = 5 \times 9 = 45$$

A farmer has **80 chickens and rabbits**. There are total of **250 legs**.
How many chickens are there? How many rabbits are there ?

Assume all were chickens

$$80 \times 2 = 160$$

$$250 - 160 = 90$$

There would be a shortage of 90 legs.
The rabbits were counted as chickens.

$$4 - 2 = 2$$

$$90 \div 2 = 45 \text{ (chickens)}$$

$$80 - 45 = 35 \text{ (rabbits)}$$

There were 35 chickens and 45 rabbits.

Assume all were rabbits

$$80 \times 4 = 320$$

$$320 - 250 = 70$$

There would be an excess of 70 legs.
The chickens were counted as rabbits.

$$4 - 2 = 2$$

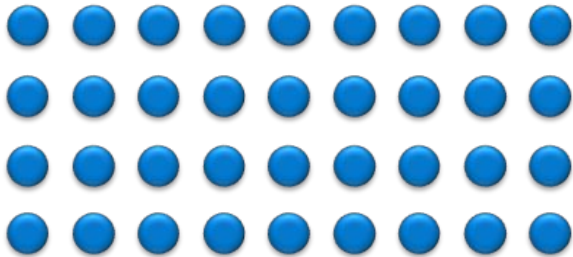
$$70 \div 2 = 35 \text{ (chickens)}$$

$$80 - 35 = 45 \text{ (rabbits)}$$

There were 35 chickens and 45 rabbits.

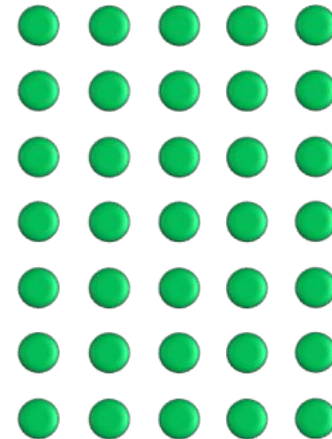
Multiple and Factors

At the sports meeting, students from two classes formed the following two formations. Calculate how many students are in each class?



$$9 \times 4 = 36$$

36 is a **multiple** of 9 and 4.
9 and 4 are **factors** of 36.



$$5 \times 7 = 35$$

35 is a **multiple** of 5 and 7.
5 and 7 are **factors** of 35.

List all the prime numbers from 1 to 100

PRIME NUMBERS

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Prime factorization is the process where a number is expressed as the product of two or more prime numbers .

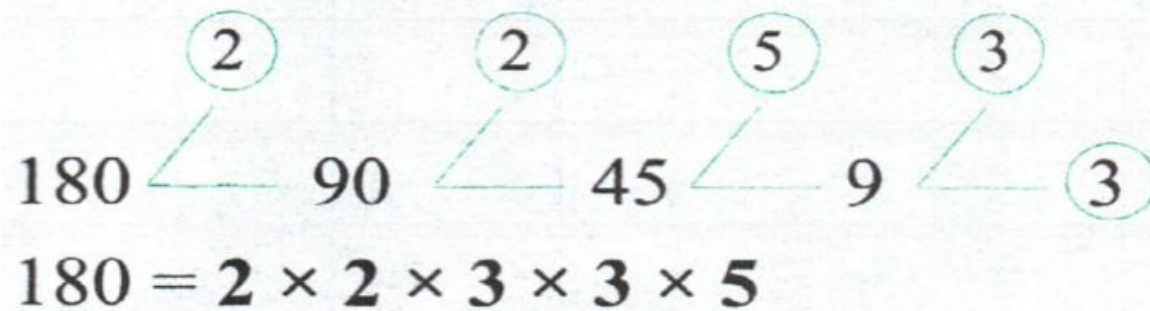


Diagram illustrating the prime factorization of 180. The number 180 is divided by 2 to get 90, which is then divided by 2 to get 45. 45 is divided by 5 to get 9, and 9 is divided by 3 to get 3. The prime factors are 2, 2, 3, 3, and 5.

$$180 = 2 \times 2 \times 3 \times 3 \times 5$$

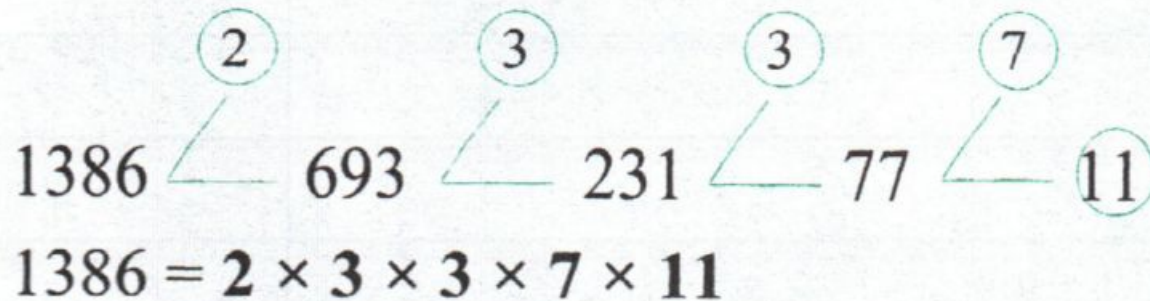


Diagram illustrating the prime factorization of 1386. The number 1386 is divided by 2 to get 693, which is then divided by 3 to get 231. 231 is divided by 3 to get 77, and 77 is divided by 7 to get 11. The prime factors are 2, 3, 3, 7, and 11.

$$1386 = 2 \times 3 \times 3 \times 7 \times 11$$

Fraction s

Conclusion

- A number composed of an integer and a fraction is called a **mixed number**; a fraction where the numerator is smaller than the denominator is called a **proper fraction**; a fraction where the numerator is greater than or equal to the denominator is called an **improper fraction**.
 - **Proper fractions** are all less than 1, and **improper fractions** are all equal to 1 or greater than 1.
-

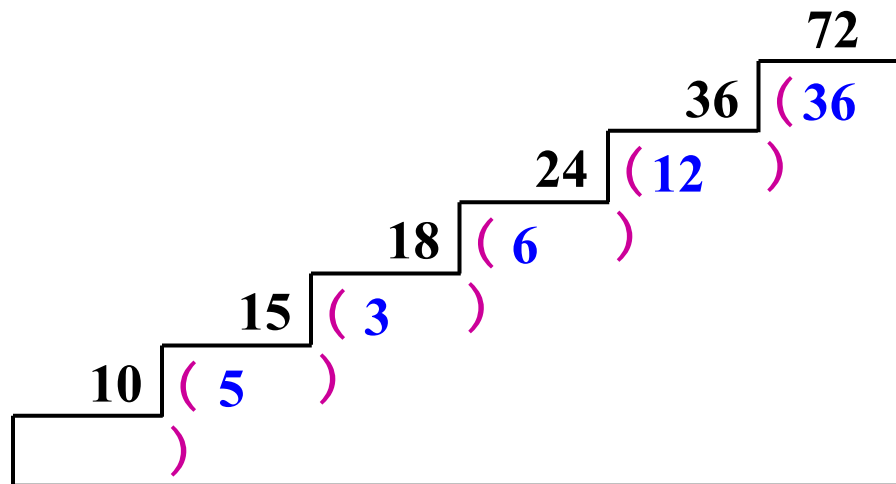
$$1\frac{1}{2} = \frac{2}{2} + \frac{1}{2} = \frac{3}{2}$$

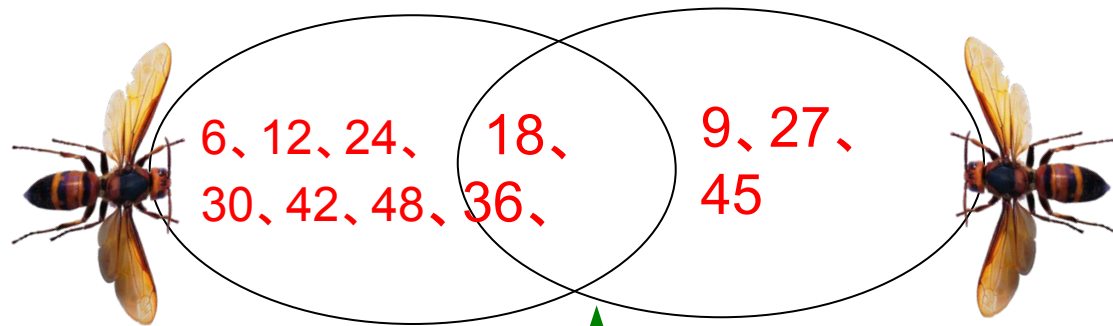
$$\frac{11}{2} = \frac{10}{2} + \frac{1}{2} = 5 + \frac{1}{2} = 5\frac{1}{2}$$

The common factors of several numbers are their common factors, and the largest one among them is their **greatest common factor**.

The common multiples of several numbers are their multiples, and the smallest one is their **least common multiple**.

Write down the greatest common factor in().





Multiples of 6 within
50

Multiples of 9 within
50

Common multiples of 6 and 9 within 50



What is their least common multiple?

18

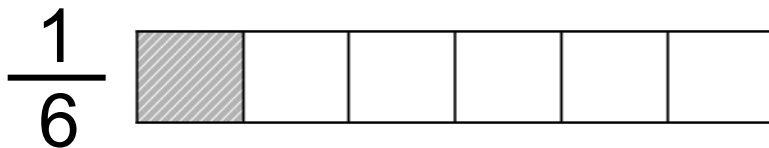
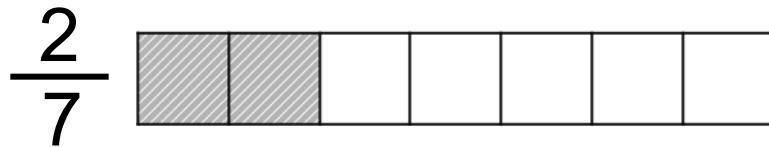
When adding or subtracting fractions with different denominators, first find a common denominator, then apply the rules of addition or subtraction for fractions with the same denominator. Simplify the result , if possible, to its simplest form.

Compare the fractions

$$\frac{2}{7} > \frac{1}{6}$$

1

Find common denominator



$$\frac{2}{7} > \frac{1}{6}$$

$$\frac{2}{7} = \frac{12}{42} \quad \frac{1}{6} = \frac{7}{42}$$

$$\frac{12}{42} > \frac{7}{42}$$

$$\frac{2}{7} > \frac{1}{6}$$

Compare the fractions

2

Find common numerator

$$\frac{2}{7} > \frac{1}{6}$$

$$\frac{1}{6} = \frac{2}{12}$$

$$\frac{2}{7} > \frac{2}{12}$$

$$\frac{2}{7} > \frac{1}{6}$$

Compare the fractions

3

Compare reciprocals

$$\frac{2}{7} > \frac{1}{6}$$

$$\begin{array}{l} \frac{7}{2} = 3 \frac{1}{2} \\ \frac{6}{1} = 6 \end{array} \left. \vphantom{\begin{array}{l} \frac{7}{2} \\ \frac{6}{1} \end{array}} \right\} \begin{array}{l} \longrightarrow \frac{7}{2} < \frac{6}{1} \\ \longrightarrow \frac{2}{7} > \frac{1}{6} \end{array}$$

4

Cross multiplication

Multiply the numerator of the first fraction by the denominator of the second fraction to obtain the relative value of the first fraction. Multiply the numerator of the second fraction by the denominator of the first fraction to obtain the relative value of the second fraction. The fraction with the larger relative value is the larger fraction.

Compare the fractions

$$\frac{7}{12} > \frac{5}{9}$$

$$\begin{array}{l} \mathbf{7 \times 9 = 63} \\ \mathbf{12 \times 5 = 60} \end{array} \left. \vphantom{\begin{array}{l} 7 \times 9 = 63 \\ 12 \times 5 = 60 \end{array}} \right\} \longrightarrow \frac{7}{12} > \frac{5}{9}$$

5

Fraction Simplification

Before comparing two fractions, simplify both fractions first, and then proceed with the comparison.

Compare the fractions

$$\frac{74}{158} < \frac{148}{259}$$

$$\frac{74}{158} = \frac{2}{5}$$

$$\frac{148}{259} = \frac{4}{7}$$

$$\left. \begin{array}{l} \frac{74}{158} = \frac{2}{5} \\ \frac{148}{259} = \frac{4}{7} \end{array} \right\} \Rightarrow \frac{2}{5} < \frac{4}{7}$$

Compare the fractions

6

Build a bridge

$$\frac{23}{77} < \frac{24}{75}$$

$$\begin{array}{rcl} \frac{23}{77} & < & \frac{24}{77} \\ \frac{24}{75} & > & \frac{24}{77} \end{array} \left. \vphantom{\begin{array}{rcl} \frac{23}{77} & < & \frac{24}{77} \\ \frac{24}{75} & > & \frac{24}{77} \end{array}} \right\} \rightarrow \frac{23}{77} < \frac{24}{75}$$

7

Compare the fractions

Convert into decimals

$$\frac{3}{4} > \frac{7}{10}$$

$$\frac{3}{4} = 0.75$$

$$\frac{7}{10} = 0.7$$

Compare the fractions

8

Find the difference with one

$$\frac{5}{6} > \frac{3}{4}$$

$$\begin{aligned}\frac{5}{6} &= 1 - \frac{1}{6} \\ \frac{3}{4} &= 1 - \frac{1}{4}\end{aligned}$$



$$\frac{1}{6} < \frac{1}{4}$$



$$\frac{5}{6} > \frac{3}{4}$$

9

Compare the fractions

Sugar water theory

Sugar=29 gram

$$\frac{29}{40} < \frac{39}{50}$$

Sugar + water=40 gram

$$\frac{39}{50} = \frac{29+10}{40+10}$$

Add 10 more gram
to the solution

Sugar water becomes sweeter as more sugar is added.

Find the number of factors

Step 1: Prime

Factorization
Express the number as a product of its prime factors.

$$n = p_1^a \times p_2^b \times p_3^c \times \dots$$

Step 2: Apply the

Formula

$$\text{Number of factors} = (a + 1) \times (b + 1) \times (c + 1) \times \dots$$

If the prime factorization of a natural number is $7 \times 13^{14} \times 19^2 \times 23$. How many factors does this number have ?

$$(1+1) \times (14+1) \times (2+1) \times (1+1) = 180$$

Find the number of factors of 2145

Prime Factorization of 2145 $2145 = 3^1 \times 5^1 \times 11^1 \times 13^1$

Number of
Factors:

$$(1 + 1) \times (1 + 1) \times (1 + 1) \times (1 + 1) = 2 \times 2 \times 2 \times 2 = 16$$

Find the number of even or odd factors

How many even factors does 3600 have ?

$$3600 = 2^4 \times 3^2 \times 5^2$$

An even factor must include at least one factor of 2. Therefore, the exponent of 2 in the factor must be at least 1. The exponents of 3 and 5 can range from 0 up to their maximum in the prime factorization.

- For the prime number 2, the exponent can be 1, 2, 3, or 4 (4 choices).
- For the prime number 3, the exponent can be 0, 1, or 2 (3 choices).
- For the prime number 5, the exponent can be 0, 1, or 2 (3 choices).

3600 has $4 \times 3 \times 3 = 36$ even factors

Find the number of even or odd factors

How many odd factors does 3600 have ?

$$3600 = 2^4 \times 3^2 \times 5^2$$

Since odd factors cannot include the prime number 2, we ignore the 2^4 part of the factorization.

- The exponent of 3 is 2, so we have $2+1=3$ choices.
- The exponent of 5 is 2, so we have $2+1=3$ choices.

3600 has $3 \times 3 = 9$ odd factors

Find the **Greatest Common Factor (GCF)** and **Least Common Multiple (LCM)** of numbers using prime factorization

Find the GCF of 360 and 700

Prime factorization of 360: $360 = 2^3 \times 3^2 \times 5^1$

Prime factorization of 700: $700 = 2^2 \times 5^2 \times 7^1$

The GCF is the product of the **lowest power of common prime factors**.

- The common primes of 360 and 700 are 2 and 5.

Take the lowest power of each common prime and multiply them together

$$\text{GCF of 360 and 700} = 2^2 \times 5 = 4 \times 5 = 20$$

Find the LCM of 360 and 700

Prime factorization of 360: $360 = 2^3 \times 3^2 \times 5^1$

Prime factorization of 700: $700 = 2^2 \times 5^2 \times 7^1$

The LCM is the product of the **highest power of all prime factors**.

- The primes of 360 and 700 are 2, 3, 5, 7.

Take the highest power of each prime:

1. For 2: the highest power is 2^3 (from 360).
2. For 3: the highest power is 3^2 (from 360).
3. For 5: the highest power is 5^2 (from 700).
4. For 7: the highest power is 7^1 (from 700).

Multiply them together:

$$\text{LCM of 360 and 700} = 2^3 \times 3^2 \times 5^2 \times 7^1 = 8 \times 9 \times 25 \times 7 = 12600$$

Euclidean Algorithm

The **Euclidean algorithm** is a method for finding the **greatest common factor (GCF)** of two numbers. It is based on the principle that the GCF of two numbers also divides their difference. The process involves repeated division, where we replace the larger number with the remainder of the division until the remainder is zero. The last non-zero remainder is the GCF.

Let's find the GCD of two numbers, say a and b , where $a > b$.

1. Divide a by b and find the remainder r :

2. $a = b \cdot q + r$:

1. q is the quotient.

2. r is the remainder ($0 \leq r < b$).

3. Replace a with b and replace b with r .

4. Repeat the process until the remainder $r=0$. The last non-zero remainder is the GCF.

Example

Find the GCF of **2205** and **2475** using the Euclidean algorithm.

Iteration 1:

- $a = 2475, b = 2205$.
- Divide 2475 by 2205:

$$2475 = 2205 \cdot 1 + 270$$

- Remainder $r = 270$.

Iteration 2:

- Replace $a = 2205, b = 270$.
- Divide 2205 by 270:

$$2205 = 270 \cdot 8 + 45$$

- Remainder $r = 45$.

Example

Find the GCF of **2205** and **2475** using the Euclidean algorithm.

Iteration 3:

- Replace $a = 270$, $b = 45$.
- Divide 270 by 45:

$$270 = 45 \cdot 6 + 0$$

- Remainder $r = 0$.

The last non-zero remainder is 45.

So, $\text{GCF}(2205, 2475) = 45$.

Given that the greatest common factor (GCF) of two numbers is 6 and their least common multiple (LCM) is 72, what are these two numbers?

Let the two numbers be $6A$ and $6B$

$$6A \times 6B = \text{GCF} \times \text{LCM} = 6 \times 72 \Rightarrow A \times B = 12$$

A and B are coprime

}

$\Rightarrow$

$A, B = 1 \text{ and } 12$

$A, B = 3 \text{ and } 4$

}

$\Rightarrow$ the two numbers $= 1 \times 6$ and 12×6
the two numbers $= 3 \times 6$ and 4×6

The unique pairs of numbers that satisfy the given conditions are:
 $(6, 72)$ and $(18, 24)$.

$$\frac{1}{6} + \frac{1}{4} = \frac{5}{12}$$

$$\frac{5}{6} - \frac{3}{8} = \frac{7}{24}$$

$$\frac{3}{4} - \frac{7}{10} = \frac{1}{20}$$

$$\frac{1}{4} + \frac{5}{12} = \frac{2}{3}$$

$$1 - \frac{2}{5} = \frac{3}{5}$$

$$\frac{3}{14} + \frac{2}{7} = \frac{5}{14}$$

The meaning of multiplying a fraction by an integer:

Simplifying the operation of adding the same number several times.

The method of multiplying a fraction by an integer:

The product of the numerator and the integer becomes the new numerator, and the denominator remains unchanged.

Computation

$$5 \times \frac{3}{7} \boxtimes \frac{15}{7}$$

$$\frac{5}{9} \times 6 \boxtimes \frac{10}{3}$$

$$100 \times \frac{2}{5} \boxtimes 40$$

$$\frac{7}{12} \times 4 \boxtimes \frac{7}{3}$$

- Dividing a number by a fraction is equivalent to multiplying the number by the reciprocal of the fraction.
- When a number (excluding 0) is divided by a fraction, if the divisor is a proper fraction, the quotient is greater than the dividend;

If the divisor is equal to 1 or a fraction greater than 1, then the quotient is equal to or less than the dividend.

$$4 \div \frac{2}{11} = 22$$

$$3 \div \frac{9}{10} \boxtimes \frac{10}{3}$$

$$10 \div \frac{10}{13} = 13$$

$$12 \div \frac{1}{3} = 36$$

$$\frac{1}{3} \div 12 \boxtimes \frac{1}{36}$$

$$\frac{1}{3} \div \frac{1}{12} = 4$$

Decimal s

The method of interchange

Fractions with a denominator of 10 can be represented with one decimal place, fractions with a denominator of 100 can be represented with two decimal places, and so on.

Convert the following fractions into decimals

$$\frac{3}{100} = 0.03$$

$$\frac{156}{1000} = 0.156$$

$$\frac{37}{1000} = 0.037$$

$$\frac{9}{10} = 0.9$$

$$\frac{8}{100} = 0.08$$

$$\frac{97}{100} = 0.97$$

The method of comparing decimal numbers:

First look at the **integer part**, the number with the larger integer part is larger;

if the integer parts are the same, then look at the decimal part, first compare the **tenths place**, then compare the **hundredths place**...

Arrange the following numbers in ascending order

1. 0.52 0.519 0.543 0.534 0.54

(0.51) < (0.52) < (0.534) < (0.54) < (0.543)
9

2. 1.011 1.001 1.1 1.101

(1.00) < (1.01) < (1.1) < (1.101)
1 1 1 1

When performing step-by-step calculations in columnar form, pay attention to aligning the decimal points.

When conducting mixed operations involving decimal addition and subtraction, prioritize calculations within parentheses, and for those without parentheses, follow the order from left to right.

The laws of addition and subtraction for integers also apply to decimals, simplify calculations whenever possible.

$3.7 + 2.36 = 6.06$

$15.4 + 2.97 = 18.37$

$11.65 - 7.39 = 4.26$

$10 - 8.75 = 1.25$

$5.94 + 10.7 = 16.64$

$9.82 - 6.18 = 3.64$

